

Relativistic (Episode 1) – Hamiltonian Mechanics

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Part I

Formulation

I.1 The Basic Variables

As we know so far, we have Lagrangian Mechanics, symbolizing $\mathcal{L}$ for most systems of interest, it is just a difference between kinetic and potential energy $\mathcal{L} = T - U$.

But if we got the Lagrangian function with n generalized coordinates, then perhaps we get $q_1, \dots, q_n$ coordinates and n generalized velocities (or the derivative of the coordinates $\dot{q}_1, \dots, \dot{q}_n$):

$$\mathcal{L} = \mathcal{L}(q_1, \dots, q_n; \dot{q}_1, \dots, \dot{q}_n; t) = T - U \quad (1)$$

Configuration space: We specify a position or “configuration” of the system by using n -coordinates $(q_1, \dots, q_n)$

State space: With specify a set of initial conditions by using $2n$ -coordinates $(\dot{q}_1, \dots, \dot{q}_n)$ to define a point and specify a set of initial conditions that determine a unique solution of n sets of second-order differential equations of motion:

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \quad (i = 1, \dots, n) \quad (2)$$

I.2 Legendre Transformation

Given a function $f(x, y)$ – A two-variable function, we get:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \quad (3)$$

We introduce a new variable conjugate to x ,

$$w = \frac{\partial f}{\partial x}, \quad (4)$$

and assume that this relation can be inverted to write $x = x(w, y)$. The Legendre transform of f with respect to x is then the function

$$g(w, y) = wx - f(x, y). \quad (5)$$

Its total differential is

$$dg = x dw + w dx - \left(\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \right) = x dw - \frac{\partial f}{\partial y} dy, \quad (6)$$

where the two terms proportional to dx cancel because $w = \partial f / \partial x$. Therefore,

$$\frac{\partial g}{\partial w} = x, \quad \frac{\partial g}{\partial y} = -\frac{\partial f}{\partial y}. \quad (7)$$

Thus, the Legendre transformation replaces the independent variable x by its conjugate variable w . It is locally invertible whenever $\partial^2 f / \partial x^2 \neq 0$.

For a system with n generalized coordinates, we perform a Legendre transformation of the Lagrangian with respect to all generalized velocities. The conjugate momenta are

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}, \quad i = 1, \dots, n, \quad (8)$$

and the resulting function is the **Hamiltonian**

$$\mathcal{H}(q_1, \dots, q_n; p_1, \dots, p_n; t) = \sum_{i=1}^n p_i \dot{q}_i - \mathcal{L}(q_i, \dot{q}_i; t). \quad (9)$$

Here the velocities $\dot{q}_i$ must be expressed as functions of (q_i, p_i, t) . This is possible for a regular Lagrangian, for which

$$\det\left(\frac{\partial^2 \mathcal{L}}{\partial \dot{q}_i \partial \dot{q}_j}\right) \neq 0. \quad (10)$$

Taking the differential of the Hamiltonian gives

$$d\mathcal{H} = \sum_{i=1}^n (p_i d\dot{q}_i + \dot{q}_i dp_i) - d\mathcal{L} = \sum_{i=1}^n \left(\dot{q}_i dp_i - \frac{\partial \mathcal{L}}{\partial q_i} dq_i \right) - \frac{\partial \mathcal{L}}{\partial t} dt. \quad (11)$$

Using the Euler-Lagrange equations, $\partial \mathcal{L} / \partial q_i = \dot{p}_i$, we obtain

$$d\mathcal{H} = \sum_{i=1}^n (\dot{q}_i dp_i - \dot{p}_i dq_i) - \frac{\partial \mathcal{L}}{\partial t} dt. \quad (12)$$

On the other hand, because $\mathcal{H} = \mathcal{H}(q_i, p_i, t)$,

$$d\mathcal{H} = \sum_{i=1}^n \left(\frac{\partial \mathcal{H}}{\partial q_i} dq_i + \frac{\partial \mathcal{H}}{\partial p_i} dp_i \right) + \frac{\partial \mathcal{H}}{\partial t} dt. \quad (13)$$

Comparing coefficients yields **Hamilton's equations**:

$$\boxed{\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}}, \quad \boxed{\dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i}}, \quad \frac{\partial \mathcal{H}}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t}. \quad (14)$$

For example, if

$$\mathcal{L}(q, \dot{q}) = \frac{1}{2} m \dot{q}^2 - U(q), \quad (15)$$

then $p = m\dot{q}$, so $\dot{q} = p/m$, and

$$\mathcal{H}(q, p) = p\dot{q} - \mathcal{L} = \frac{p^2}{2m} + U(q). \quad (16)$$

In this common case, the Hamiltonian is the total mechanical energy $T + U$.

Part II

Hamiltonian

II.1 Hamiltonian for 1D System

Consider a system with one degree of freedom, described by a generalized coordinate q and a Lagrangian

$$\mathcal{L} = \mathcal{L}(q, \dot{q}; t). \quad (17)$$

Examples include a plane pendulum, for which $q \equiv \phi$, and a bead constrained to move along a stationary wire. The conjugate momentum is

$$p = \frac{\partial \mathcal{L}}{\partial \dot{q}}. \quad (18)$$

If this relation can be inverted to obtain $\dot{q} = \dot{q}(q, p; t)$, then the Hamiltonian is

$$\mathcal{H}(q, p; t) = p\dot{q} - \mathcal{L}(q, \dot{q}; t). \quad (19)$$

The state of the system is represented by a point (q, p) in the two-dimensional **phase space**. Its time evolution is governed by

$$\boxed{\dot{q} = \frac{\partial \mathcal{H}}{\partial p}}, \quad \boxed{\dot{p} = -\frac{\partial \mathcal{H}}{\partial q}}. \quad (20)$$

These two first-order differential equations are equivalent to the single second-order Euler–Lagrange equation. For a large class of one-dimensional systems, the Lagrangian has the form

$$\mathcal{L}(q, \dot{q}) = \frac{1}{2}M(q)\dot{q}^2 - U(q), \quad (21)$$

where $M(q) > 0$ acts as a coordinate-dependent effective mass. Hence

$$p = M(q)\dot{q}, \quad \dot{q} = \frac{p}{M(q)}, \quad (22)$$

and the Hamiltonian becomes

$$\mathcal{H}(q, p) = \frac{p^2}{2M(q)} + U(q). \quad (23)$$

Hamilton's equations are therefore

$$\dot{q} = \frac{p}{M(q)}, \quad \dot{p} = \frac{p^2 M'(q)}{2M(q)^2} - U'(q). \quad (24)$$

If M is constant, these equations reduce to

$$\dot{q} = \frac{p}{M}, \quad \dot{p} = -U'(q), \quad (25)$$

which combine to give Newton's equation

$$M\ddot{q} = -U'(q). \quad (26)$$

Example 1: Plane pendulum

For a bob of mass m , attached to a massless rod of length ℓ , let ϕ be the angle measured from the downward vertical. Its kinetic and potential energies are

$$T = \frac{1}{2}m\ell^2\dot{\phi}^2, \quad U = mg\ell(1 - \cos \phi). \quad (27)$$

Thus,

$$\mathcal{L}(\phi, \dot{\phi}) = \frac{1}{2}m\ell^2\dot{\phi}^2 - mg\ell(1 - \cos \phi). \quad (28)$$

The momentum conjugate to ϕ is the angular momentum

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = m\ell^2\dot{\phi}, \quad \dot{\phi} = \frac{p_\phi}{m\ell^2}. \quad (29)$$

Therefore,

$$\mathcal{H}(\phi, p_\phi) = \frac{p_\phi^2}{2m\ell^2} + mg\ell(1 - \cos \phi). \quad (30)$$

Hamilton's equations give

$$\dot{\phi} = \frac{p_\phi}{m\ell^2}, \quad \dot{p}_\phi = -mg\ell \sin \phi. \quad (31)$$

Combining them yields the familiar pendulum equation

$$\ddot{\phi} + \frac{g}{\ell} \sin \phi = 0. \quad (32)$$

Example 2: Bead on a stationary wire

Let the shape of the wire be described by a position vector $\mathbf{r}(q)$, where q parametrizes distance along the wire. The velocity of a bead of mass m is

$$\dot{\mathbf{r}} = \frac{d\mathbf{r}}{dq} \dot{q}. \quad (33)$$

Defining

$$h(q) = \left| \frac{d\mathbf{r}}{dq} \right|, \quad (34)$$

the Lagrangian is

$$\mathcal{L}(q, \dot{q}) = \frac{1}{2} m h(q)^2 \dot{q}^2 - U(q). \quad (35)$$

The conjugate momentum and Hamiltonian are then

$$p = m h(q)^2 \dot{q}, \quad \boxed{\mathcal{H}(q, p) = \frac{p^2}{2m h(q)^2} + U(q)}. \quad (36)$$

If $q = s$ is chosen to be the arc length along the wire, then $h(s) = 1$ and the result simplifies to

$$\mathcal{H}(s, p_s) = \frac{p_s^2}{2m} + U(s). \quad (37)$$

Finally, along any Hamiltonian trajectory,

$$\frac{d\mathcal{H}}{dt} = \frac{\partial \mathcal{H}}{\partial q} \dot{q} + \frac{\partial \mathcal{H}}{\partial p} \dot{p} + \frac{\partial \mathcal{H}}{\partial t} = \frac{\partial \mathcal{H}}{\partial t}. \quad (38)$$

Consequently, if the Hamiltonian has no explicit time dependence, then $\mathcal{H}$ is conserved.

II.2 Hamilton–Jacobi Equation

Hamilton’s equations determine the trajectory $(q(t), p(t))$ through phase space. An alternative formulation seeks a single function whose derivatives generate the same motion. This function is called **Hamilton’s principal function**,

$$S(q, t) = \int_{t_0}^t \mathcal{L}(q, \dot{q}; t') dt', \quad (39)$$

where the integral is evaluated along the classical path ending at the point (q, t) . A variation of the endpoint gives

$$dS = p dq - \mathcal{H} dt. \quad (40)$$

Comparing this expression with

$$dS = \frac{\partial S}{\partial q} dq + \frac{\partial S}{\partial t} dt \quad (41)$$

shows that

$$p = \frac{\partial S}{\partial q}, \quad \mathcal{H} = -\frac{\partial S}{\partial t}. \quad (42)$$

Substituting the first relation into the second gives the **Hamilton–Jacobi equation**

$$\boxed{\mathcal{H}\left(q, \frac{\partial S}{\partial q}; t\right) + \frac{\partial S}{\partial t} = 0}. \quad (43)$$

For a system with n degrees of freedom, its general form is

$$\boxed{\mathcal{H}\left(q_1, \dots, q_n; \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}; t\right) + \frac{\partial S}{\partial t} = 0}. \quad (44)$$

Thus, Hamilton's $2n$ first-order ordinary differential equations are replaced by one first-order partial differential equation.

If the Hamiltonian has no explicit time dependence, we may separate the variables by writing

$$S(q, t) = W(q) - Et, \quad (45)$$

where E is the conserved energy and $W(q)$ is called Hamilton's **characteristic function**. The Hamilton–Jacobi equation then becomes

$$\mathcal{H}\left(q, \frac{dW}{dq}\right) = E, \quad p = \frac{dW}{dq}. \quad (46)$$

For the one-dimensional Hamiltonian

$$\mathcal{H}(q, p) = \frac{p^2}{2M(q)} + U(q), \quad (47)$$

we obtain

$$\frac{dW}{dq} = p(q) = \pm \sqrt{2M(q)[E - U(q)]}, \quad (48)$$

and hence

$$W(q) = \pm \int^q \sqrt{2M(q')[E - U(q')]} dq'. \quad (49)$$

The motion is restricted to the classically allowed region $E \geq U(q)$.

Example: Free particle

For a free particle,

$$\mathcal{H} = \frac{p^2}{2m}, \quad (50)$$

and a complete solution of the Hamilton–Jacobi equation is

$$S(q, t; p) = pq - \frac{p^2}{2m}t. \quad (51)$$

Differentiating with respect to the constant p gives another constant,

$$\beta = \frac{\partial S}{\partial p} = q - \frac{p}{m}t. \quad (52)$$

Taking $\beta = q_0$, the trajectory is

$$q(t) = q_0 + \frac{p}{m}t, \quad (53)$$

which is precisely the uniform motion obtained from Hamilton's equations.

II.3 Poisson Brackets

Let $F(q_i, p_i, t)$ and $G(q_i, p_i, t)$ be two functions on phase space. Their **Poisson bracket** is defined by

$$\{F, G\} = \sum_{i=1}^n \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right). \quad (54)$$

The fundamental Poisson brackets of the canonical coordinates are

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}, \quad (55)$$

where δ_{ij} is the Kronecker delta.

Hamilton's equations can now be written compactly as

$$\dot{q}_i = \{q_i, \mathcal{H}\}, \quad \dot{p}_i = \{p_i, \mathcal{H}\}. \quad (56)$$

More generally, the total time derivative of any phase-space function is

$$\frac{dF}{dt} = \sum_{i=1}^n \left(\frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial p_i} \dot{p}_i \right) + \frac{\partial F}{\partial t} \quad (57)$$

$$= \boxed{\{F, \mathcal{H}\} + \frac{\partial F}{\partial t}}. \quad (58)$$

Therefore, a function F is conserved when

$$\{F, \mathcal{H}\} + \frac{\partial F}{\partial t} = 0. \quad (59)$$

In particular, if F has no explicit time dependence, then $\{F, \mathcal{H}\} = 0$ is the condition for conservation. Since every function has a vanishing Poisson bracket with itself,

$$\{\mathcal{H}, \mathcal{H}\} = 0, \quad (60)$$

an explicitly time-independent Hamiltonian is conserved. The Poisson bracket satisfies the following useful identities:

$$\{F, G\} = -\{G, F\} \quad (\text{antisymmetry}), \quad (61)$$

$$\{F, G + K\} = \{F, G\} + \{F, K\} \quad (\text{linearity}), \quad (62)$$

$$\{F, GK\} = \{F, G\}K + G\{F, K\} \quad (\text{product rule}), \quad (63)$$

$$\{F, \{G, K\}\} + \{G, \{K, F\}\} + \{K, \{F, G\}\} = 0 \quad (\text{Jacobi identity}). \quad (64)$$

An important consequence is Poisson's theorem: if F and G are constants of motion with no explicit time dependence, then their Poisson bracket $\{F, G\}$ is also a constant of motion.

Example: Harmonic oscillator

For

$$\mathcal{H}(q, p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2, \quad (65)$$

the bracket equations give

$$\dot{q} = \{q, \mathcal{H}\} = \frac{p}{m}, \quad \dot{p} = \{p, \mathcal{H}\} = -m \omega^2 q. \quad (66)$$

Combining these equations yields

$$\ddot{q} + \omega^2 q = 0. \quad (67)$$

Thus, Poisson brackets reproduce the equations of motion while also making the algebraic structure of Hamiltonian mechanics explicit.

Part III

Relations with Classical Statistical Mechanics

III.1 Phase Space

Classical statistical mechanics applies Hamiltonian mechanics to systems containing a very large number of degrees of freedom. Instead of following one exact trajectory, we describe an ensemble of possible microscopic states by a probability distribution on phase space.

For N point particles in three dimensions, a microscopic state is specified by the $3N$ Cartesian coordinates and $3N$ conjugate momenta,

$$\mathbf{q} = (q_1, \dots, q_N), \quad \mathbf{p} = (p_1, \dots, p_N). \quad (68)$$

The corresponding phase space therefore has dimension $6N$. More generally, for f degrees of freedom, a phase point is written as

$$z = (q_1, \dots, q_f, p_1, \dots, p_f). \quad (69)$$

Hamilton's equations define a velocity field on this space:

$$\dot{z} = (\dot{q}_1, \dots, \dot{q}_f, \dot{p}_1, \dots, \dot{p}_f). \quad (70)$$

Starting from an initial point z_0 , these equations generate a phase-space trajectory $z(t)$. Because the equations of motion have a unique solution, two distinct trajectories cannot intersect at the same time.

It is convenient to introduce the symplectic matrix

$$\mathbb{J} = \begin{pmatrix} 0 & I_f \\ -I_f & 0 \end{pmatrix}, \quad (71)$$

where I_f is the $f \times f$ identity matrix. Hamilton's equations then take the compact form

$$\dot{z} = \mathbb{J} \nabla_z \mathcal{H}. \quad (72)$$

The matrix $\mathbb{J}$ encodes the pairing of each coordinate q_i with its conjugate momentum p_i .

Energy surfaces and accessible states

If the Hamiltonian has no explicit time dependence, then the energy is conserved. A trajectory with energy E is therefore restricted to the constant-energy hypersurface

$$\mathcal{H}(q, p) = E. \quad (73)$$

For f degrees of freedom, this surface generally has dimension $2f - 1$. Additional conserved quantities, such as total momentum or angular momentum, restrict the motion to still smaller regions of phase space.

A **microstate** is one exact phase point. A **macrostate**, by contrast, is specified by a few macroscopic quantities such as energy, volume, pressure, and particle number. An enormous number of microstates may correspond to the same macrostate. Statistical mechanics assigns probabilities to these microscopic possibilities.

Phase-space volume

The natural infinitesimal volume element in phase space is

$$d\Gamma = \prod_{i=1}^f dq_i dp_i. \quad (74)$$

For N particles in three dimensions this becomes

$$d\Gamma = \prod_{a=1}^N d^3 q_a d^3 p_a. \quad (75)$$

The volume of a region R is

$$\Gamma(R) = \int_R d\Gamma. \quad (76)$$

Although classical phase space is continuous, the factor h^{3N} is often used to make the number of states dimensionless. For N identical particles, the statistical measure is conventionally written as

$$d\mu_N = \frac{d\Gamma}{N! h^{3N}}, \quad (77)$$

where the factor $N!$ avoids overcounting permutations of identical particles.

Probability density and ensemble averages

Let $\rho(q, p, t)$ be a normalized phase-space probability density:

$$\rho(q, p, t) \geq 0, \quad \int \rho(q, p, t) d\Gamma = 1. \quad (78)$$

The probability of finding the system inside a region R is

$$P(R, t) = \int_R \rho(q, p, t) d\Gamma. \quad (79)$$

For any observable $A(q, p, t)$, its ensemble average is

$$\langle A \rangle = \int A(q, p, t) \rho(q, p, t) d\Gamma. \quad (80)$$

Its statistical variance is

$$(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2. \quad (81)$$

If only a subset of the variables is observed, the unobserved variables can be integrated out. For example, a configuration-space density is obtained from

$$\rho_q(q, t) = \int \rho(q, p, t) d^f p. \quad (82)$$

Such reduced distributions contain less microscopic information than the full phase-space density.

Example: Harmonic-oscillator phase portrait

For the one-dimensional harmonic oscillator,

$$\mathcal{H}(q, p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2 = E. \quad (83)$$

The constant-energy curves are ellipses in the (q, p) plane. Their semi-axes are

$$q_{\max} = \sqrt{\frac{2E}{m\omega^2}}, \quad p_{\max} = \sqrt{2mE}. \quad (84)$$

Therefore, the phase-space area enclosed by an orbit is

$$\Gamma(E) = \pi q_{\max} p_{\max} = \frac{2\pi E}{\omega}. \quad (85)$$

The associated action variable is

$$J = \frac{1}{2\pi} \oint p dq = \frac{E}{\omega}. \quad (86)$$

This example illustrates how geometrical structures in phase space encode the dynamics.

III.2 Liouville's Theorem

Liouville's theorem states that Hamiltonian evolution preserves phase-space volume. Equivalently, an ensemble moves through phase space like an incompressible fluid: it can stretch and deform, but its local fine-grained density does not change along a trajectory.

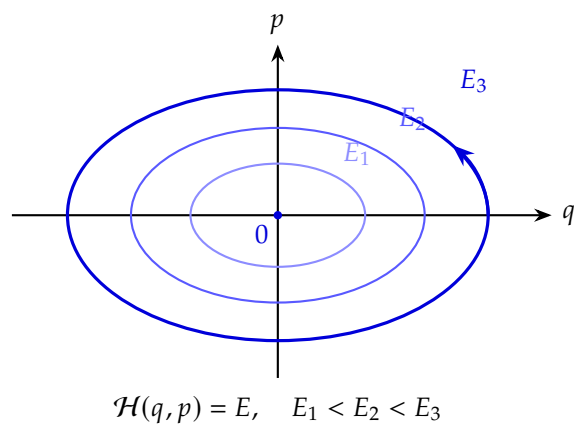


Figure 1: Phase portrait of a one-dimensional harmonic oscillator. Every ellipse is a constant-energy orbit, and the arrows indicate Hamiltonian motion around the orbit.

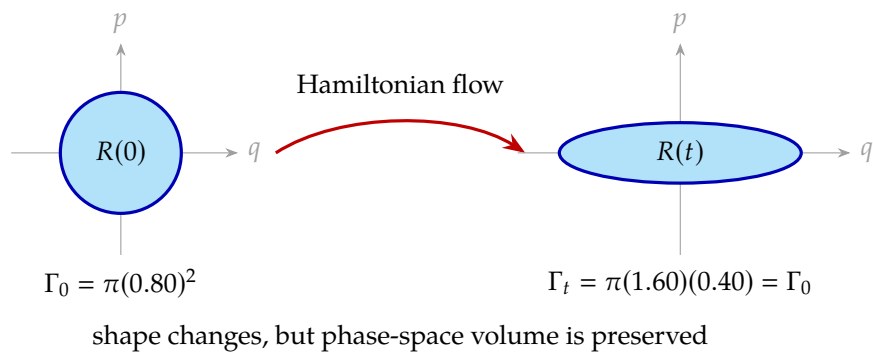


Figure 2: Two-dimensional illustration of Liouville’s theorem. An initially compact ensemble can stretch under Hamiltonian evolution, while its phase-space area remains unchanged.

Continuity equation in phase space

Conservation of probability implies the phase-space continuity equation

$$\frac{\partial \rho}{\partial t} + \sum_{i=1}^f \left[\frac{\partial}{\partial q_i} (\rho \dot{q}_i) + \frac{\partial}{\partial p_i} (\rho \dot{p}_i) \right] = 0. \quad (87)$$

Expanding the derivatives gives

$$\frac{D\rho}{Dt} + \rho \sum_{i=1}^f \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) = 0, \quad (88)$$

where

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + \sum_{i=1}^f \left(\dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} \right) \quad (89)$$

is the derivative following the Hamiltonian flow.

Using Hamilton's equations,

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i}, \quad (90)$$

the divergence of the phase-space velocity is

$$\nabla_z \cdot \dot{z} = \sum_{i=1}^f \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) \quad (91)$$

$$= \sum_{i=1}^f \left(\frac{\partial^2 \mathcal{H}}{\partial q_i \partial p_i} - \frac{\partial^2 \mathcal{H}}{\partial p_i \partial q_i} \right) = 0. \quad (92)$$

Consequently,

$$\boxed{\frac{D\rho}{Dt} = 0}. \quad (93)$$

Thus, the probability density attached to a moving phase point remains constant.

Using Poisson brackets, the same result can be written as the **Liouville equation**

$$\boxed{\frac{\partial \rho}{\partial t} + \{\rho, \mathcal{H}\} = 0}. \quad (94)$$

This equation is the classical analogue of the von Neumann equation in quantum mechanics.

Preservation of finite volumes

Let $R(0)$ be a small region of initial conditions and let $R(t)$ be the region obtained by evolving every point according to Hamilton's equations. If

$$J(t) = \det \left(\frac{\partial z(t)}{\partial z(0)} \right) \quad (95)$$

is the Jacobian of the evolution map, then

$$\frac{dJ}{dt} = J \nabla_z \cdot \dot{z} = 0. \quad (96)$$

Since $J(0) = 1$, it follows that $J(t) = 1$, and hence

$$\boxed{\int_{R(t)} d\Gamma = \int_{R(0)} d\Gamma}. \quad (97)$$

Hamiltonian flow may turn a compact region into a long and complicated filament, but it cannot change its total volume.

Stationary distributions

A distribution is stationary if

$$\frac{\partial \rho}{\partial t} = 0. \quad (98)$$

The Liouville equation then requires

$$\{\rho, \mathcal{H}\} = 0. \quad (99)$$

Any distribution depending only on constants of motion is stationary. In particular,

$$\rho(q, p) = F(\mathcal{H}(q, p)) \quad (100)$$

satisfies

$$\{\rho, \mathcal{H}\} = F'(\mathcal{H})\{\mathcal{H}, \mathcal{H}\} = 0. \quad (101)$$

The microcanonical and canonical equilibrium distributions introduced below are both of this form.

Example: Rotation of an oscillator ensemble

For the harmonic oscillator, define scaled canonical variables

$$Q = \sqrt{m\omega} q, \quad P = \frac{p}{\sqrt{m\omega}}. \quad (102)$$

The Hamiltonian becomes

$$\mathcal{H} = \frac{\omega}{2}(Q^2 + P^2), \quad (103)$$

and Hamilton's equations are

$$\dot{Q} = \omega P, \quad \dot{P} = -\omega Q. \quad (104)$$

They describe a rigid rotation in the (Q, P) plane. Any ensemble region rotates without changing its area, providing a simple visual example of Liouville's theorem.

Volume preservation is a special property of Hamiltonian systems. For comparison, a damped oscillator with $\dot{p} = -\partial\mathcal{H}/\partial q - \gamma p$ has

$$\nabla_z \cdot \dot{z} = -\gamma, \quad (105)$$

so its phase-space area contracts exponentially. Dissipative motion is therefore not described by an autonomous canonical Hamiltonian in the same phase space.

III.3 Equilibrium Ensembles

An ensemble is a hypothetical collection of many copies of a system, prepared under the same macroscopic conditions but occupying different microstates. Different physical constraints lead to different equilibrium probability distributions.

Microcanonical ensemble

The microcanonical ensemble describes an isolated system with fixed energy E , volume V , and particle number N . Define the phase-space volume below energy E by

$$\Phi(E) = \int \Theta(E - \mathcal{H}(q, p)) d\Gamma, \quad (106)$$

where Θ is the Heaviside step function. The density of states is

$$\Omega(E) = \frac{\partial \Phi}{\partial E} = \int \delta(E - \mathcal{H}(q, p)) d\Gamma. \quad (107)$$

The normalized microcanonical distribution is

$$\rho_{\text{mc}}(q, p) = \frac{\delta(E - \mathcal{H}(q, p))}{\Omega(E)}. \quad (108)$$

It assigns equal statistical weight to all accessible microstates on the energy surface. The entropy may be written, up to the conventional choice of energy width and reference cell, as

$$S(E, V, N) = k_B \ln \Omega(E, V, N) + \text{constant}. \quad (109)$$

Thermodynamic quantities follow from derivatives of the entropy, for example

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{V, N}, \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V} \right)_{E, N}. \quad (110)$$

Canonical ensemble

The canonical ensemble describes a system that can exchange energy with a large reservoir at temperature T , while V and N remain fixed. Its probability density is

$$\rho_c(q, p) = \frac{e^{-\beta \mathcal{H}(q, p)}}{Z}, \quad \beta = \frac{1}{k_B T}, \quad (111)$$

where

$$Z(T, V, N) = \int e^{-\beta \mathcal{H}(q, p)} d\mu_N \quad (112)$$

is the canonical partition function. The partition function contains all equilibrium thermodynamic information. In particular,

$$F = -k_B T \ln Z, \quad (113)$$

$$U = \langle \mathcal{H} \rangle = -\frac{\partial \ln Z}{\partial \beta}, \quad (114)$$

$$S = k_B (\ln Z + \beta U), \quad (115)$$

$$(\Delta E)^2 = \frac{\partial^2 \ln Z}{\partial \beta^2} = k_B T^2 C_V. \quad (116)$$

For macroscopic systems, $\Delta E/U$ is usually extremely small, which explains why the canonical and microcanonical ensembles give equivalent bulk predictions in the thermodynamic limit.

Example: Classical ideal gas

For N noninteracting particles of mass m in a volume V ,

$$\mathcal{H} = \sum_{a=1}^N \frac{p_a^2}{2m}. \quad (117)$$

The canonical partition function is

$$Z_N = \frac{1}{N!} \left(\frac{V}{\lambda_T^3} \right)^N, \quad \lambda_T = \frac{h}{\sqrt{2\pi m k_B T}}, \quad (118)$$

where λ_T is the thermal de Broglie wavelength. Differentiation of $\ln Z_N$ gives

$$U = \frac{3}{2} N k_B T, \quad PV = N k_B T. \quad (119)$$

Thus, the familiar ideal-gas laws follow directly from an integral over classical phase space.

Grand canonical ensemble

If the system can exchange both energy and particles with a reservoir, the temperature T , volume V , and chemical potential μ are fixed. The grand partition function is

$$\Xi(T, V, \mu) = \sum_{N=0}^{\infty} e^{\beta\mu N} Z_N(T, V). \quad (120)$$

Writing the fugacity as $z = e^{\beta\mu}$, the mean particle number is

$$\langle N \rangle = z \frac{\partial \ln \Xi}{\partial z}, \quad (121)$$

and the grand potential is

$$\mathcal{G} = -k_B T \ln \Xi = -PV. \quad (122)$$

Particle-number fluctuations are determined by

$$(\Delta N)^2 = z \frac{\partial \langle N \rangle}{\partial z}. \quad (123)$$

III.4 Equipartition Theorem

In the classical canonical ensemble, integration by parts leads to the general equipartition identity

$$\left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle = k_B T \delta_{ij}, \quad (124)$$

where x_i denotes a canonical coordinate or momentum and boundary terms are assumed to vanish. Every independent quadratic term in the Hamiltonian therefore contributes

$$\frac{1}{2} k_B T \quad (125)$$

to the mean energy.

For a free particle in three dimensions, the Hamiltonian contains three quadratic momentum terms, so

$$\left\langle \frac{\mathbf{p}^2}{2m} \right\rangle = \frac{3}{2} k_B T. \quad (126)$$

For a one-dimensional harmonic oscillator,

$$\mathcal{H} = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2, \quad (127)$$

there is one quadratic kinetic term and one quadratic potential term:

$$\left\langle \frac{p^2}{2m} \right\rangle = \left\langle \frac{1}{2} m \omega^2 q^2 \right\rangle = \frac{1}{2} k_B T. \quad (128)$$

Hence the oscillator has mean energy

$$\langle \mathcal{H} \rangle = k_B T. \quad (129)$$

Equipartition is a classical result and can fail at low temperatures, where quantum effects become important.

III.5 Entropy, Ergodicity, and Irreversibility

For a normalized phase-space distribution, the fine-grained Gibbs entropy is

$$S_G = -k_B \int \rho(q, p, t) \ln[\rho(q, p, t) \Gamma_0] d\Gamma, \quad (130)$$

where Γ_0 is a fixed reference phase-space volume inserted to make the logarithm dimensionless. Because Hamiltonian evolution preserves both ρ along trajectories and the measure $d\Gamma$, Liouville's theorem implies

$$\frac{dS_G}{dt} = 0 \quad (131)$$

for an isolated system under fine-grained evolution.

This result appears to conflict with the thermodynamic statement that entropy increases. The resolution is that macroscopic observations cannot resolve arbitrarily fine phase-space structures. Hamiltonian flow stretches and folds a distribution into thinner filaments. A **coarse-grained** description averages over small cells and discards this fine information; its entropy can increase even though the exact fine-grained entropy remains constant.

Boltzmann's entropy formula expresses the same statistical idea:

$$S_B = k_B \ln W, \quad (132)$$

where W measures the phase-space volume of microstates compatible with a given macrostate. Equilibrium corresponds to the macrostate occupying the overwhelming majority of the accessible phase-space volume.

The **ergodic hypothesis** states, roughly, that over a sufficiently long time a typical trajectory explores the accessible energy surface in a representative way. It then permits the replacement

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(q(t), p(t)) dt = \langle A \rangle_{mc}. \quad (133)$$

Ergodicity is not guaranteed for every Hamiltonian system. Integrable systems possess many constants of motion and their trajectories lie on lower-dimensional invariant tori. Chaotic systems can explore phase space more broadly, although chaos alone does not automatically prove ergodicity.

Liouville's theorem, equilibrium ensembles, and the ergodic idea together form the bridge between reversible microscopic Hamiltonian dynamics and the probabilistic laws of macroscopic thermodynamics.